Mean : $E(X) = \sum x f(x) = \frac{115}{25} = 4.6$

Variance : $V(X) = E(X^2) - \left(E(X)\right)^2 = \sum x^2 f(x) - \left(\sum x f(x)\right)^2$

$$= \frac{585}{25} - \left(\frac{115}{25}\right)^2 = 23.40 - 21.16 = 2.24$$

Therefore the mean and variance are 4.6 and 2.24 respectively. ■

Example 11.17

Two balls are chosen randomly from an urn containing 8 white and 4 black balls. Suppose that we win Rs 20 for each black ball selected and we lose Rs10 for each white ball selected. Find the expected winning amount and variance.

Solution

Let X denote the winning amount. The possible events of selection are (i) both balls are black, or (ii) one white and one black or (iii) both are white. Therefore X is a random variable that can be defined as

X (both are black balls) = ₹ 2(20) = ₹ 40

X (one black and one white ball) = ₹ 20 – ₹ 10 = ₹10

X (both are white balls) = ₹ (–20) = – ₹ 20

Therefore X takes on the values 40,10 and –20

Total number of balls $n = 12$

Total number of ways of selecting 2 balls $= \binom{12}{2} = \frac{12 \times 11}{1 \times 2} = 66$

Number of ways of selecting 2 black balls $= \binom{4}{2} = 6$

Number of ways of selecting one black ball and one white ball $= \binom{8}{1}\binom{4}{1} = 32$

Number of ways of selecting 2 white balls $= \binom{8}{2} = 28$

Values of Random Variable X	40	10	–20	Total
Number of elements in inverse images	6	32	28	66

Probability mass function is

X	40	10	–20	Total
$f(x)$	$\frac{6}{66}$	$\frac{32}{66}$	$\frac{28}{66}$	1

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